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# Overview of Internet

Dr. Swapna M P

# Contents

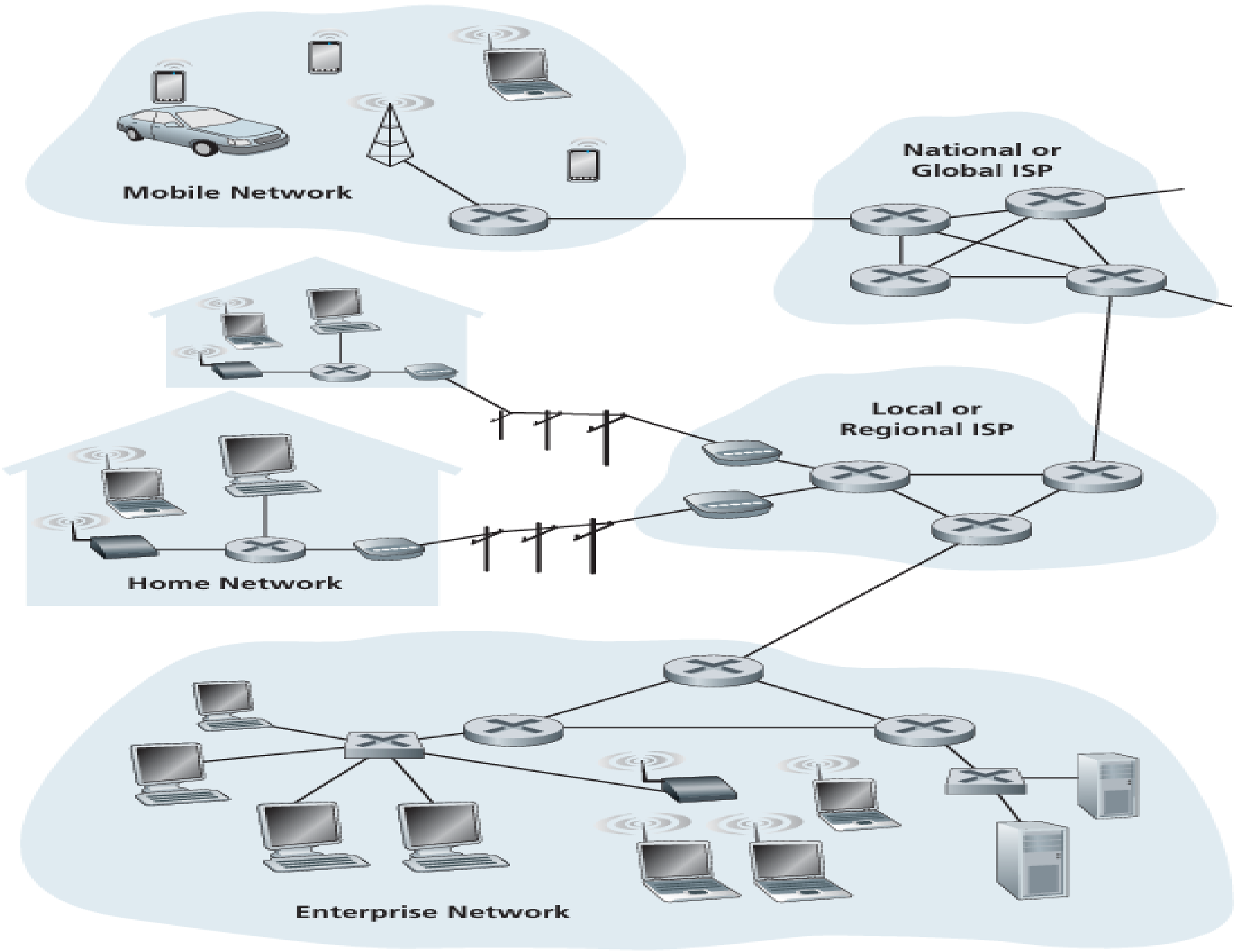
- Internet
- Protocols
- Physical Media
- Satellite Radio Channels

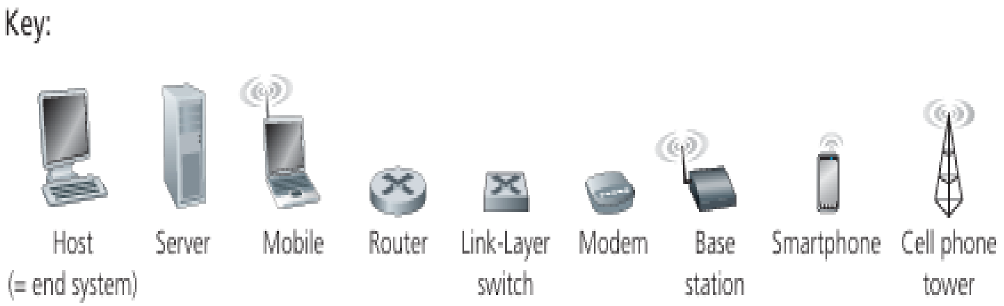
# Internet

- Global system of interconnected computer networks.

Nuts-and-Bolts view:

- Refers to the basic hardware and software components
- Hosts/end systems : all connected devices on the Internet.
- End systems are connected through a network of communication links and packet switches.
- Links exhibit diverse data transmission rates measured in bits/second.


National or
Global ISP
Mobile Network
Local or
Regional ISP
Home Network
Enterprise Network


Key:
Host
(= end system)
Server
Mobile
Router
Link-Layer
switch
Modem
Base
station
Smartphone
Cell phone
tower

- Data is transmitted through the Internet in packets.
- Packet switches forward packets towards their destinations.
- Prominent types are routers and link-layer switches.
- End systems access the Internet through Internet Service Providers (ISPs).
- Protocol : controls information exchange within the Internet.
- TCP (Transmission Control Protocol) and IP (Internet Protocol) are most Internet protocols.

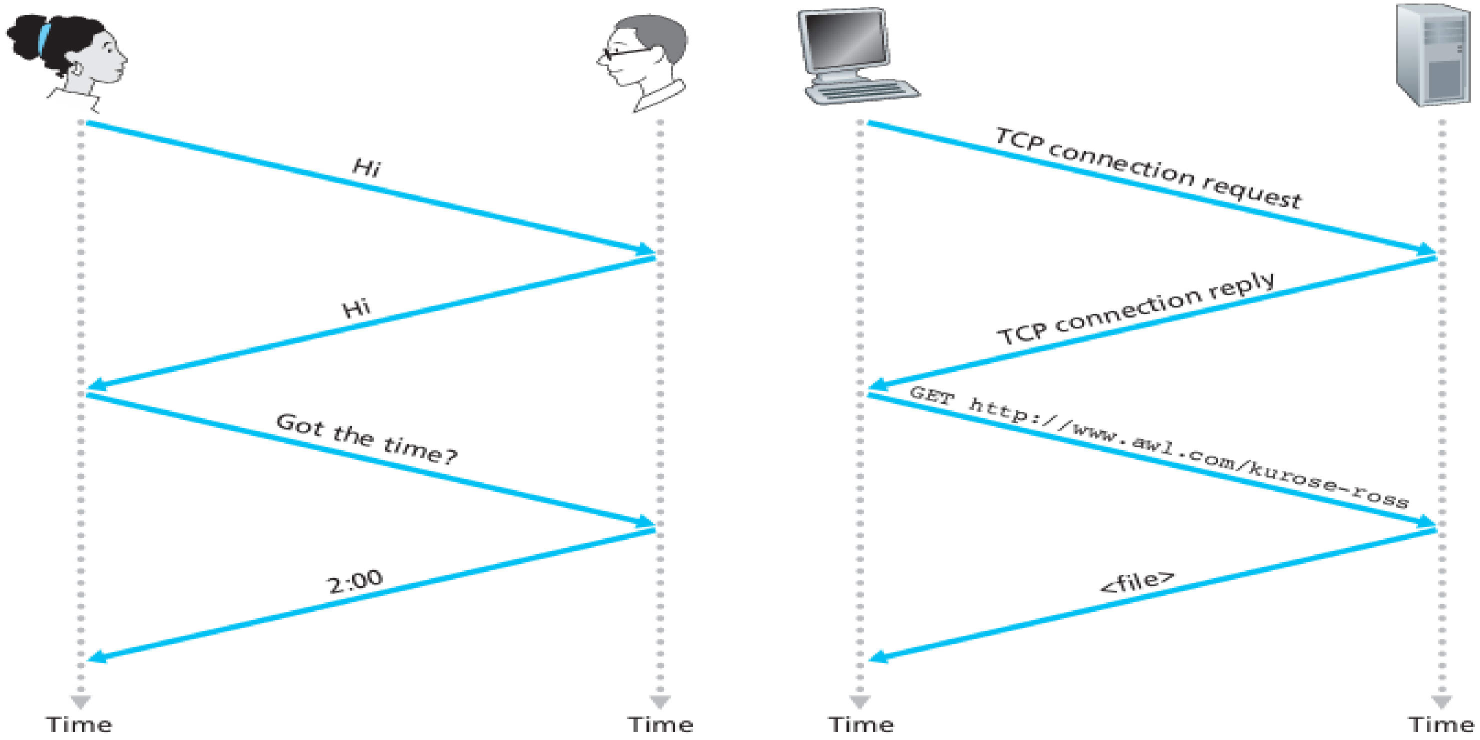
## Service View:

- Infrastructure that provides services to applications
- Eg: email, web surfing, social networks to VoIP, video streaming, and remote login.

# Protocol

- A protocol defines the format and the order of messages exchanged between two or more communicating entities, as well as the actions taken on the transmission and/or receipt of a message or other event.
- Different protocols are used to accomplish different communication tasks.

# Human and Computer Network Protocol



# Physical Media

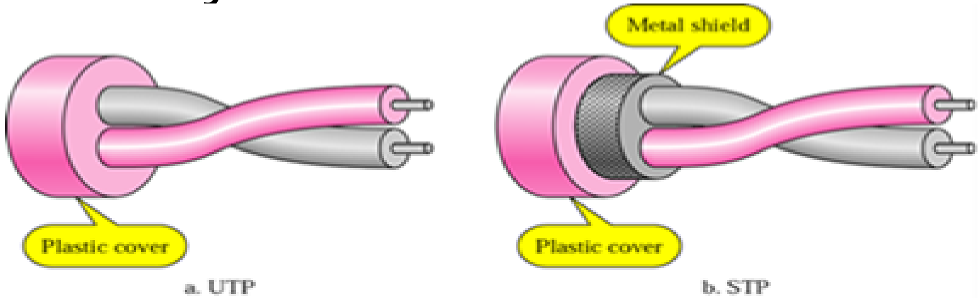
- A bit propagates between transmitter-receiver pairs.
- Each transmitter-receiver pair uses physical media.
- 2 types
- Guided Media : Signals are guided along a solid medium

  eg. Twisted-pair copper wire, Coaxial cable, Fibre-optic cable

- Unguided Media : Signals propagate freely
- eg: Wireless LAN , Digital satellite channel

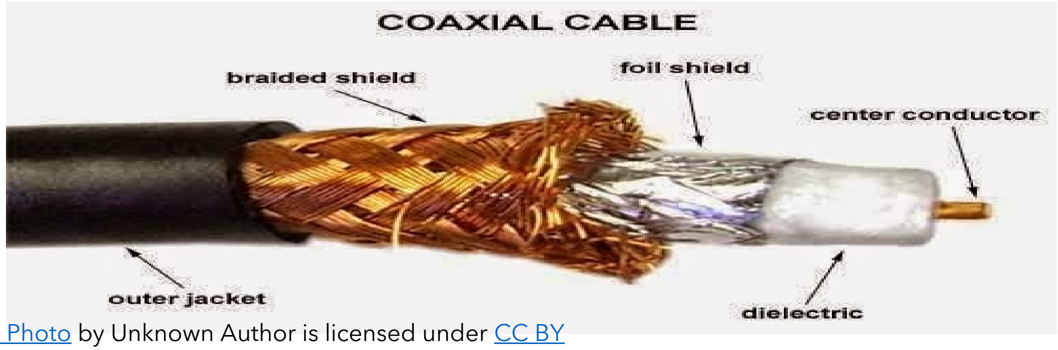
# Twisted-Pair Copper Wire

- Most widely used and cost-effective guided transmission medium
- Two insulated copper wires arranged in a regular spiral pattern
- Reduces electrical interference
- Unshielded Twisted Pair is commonly used for LANs.



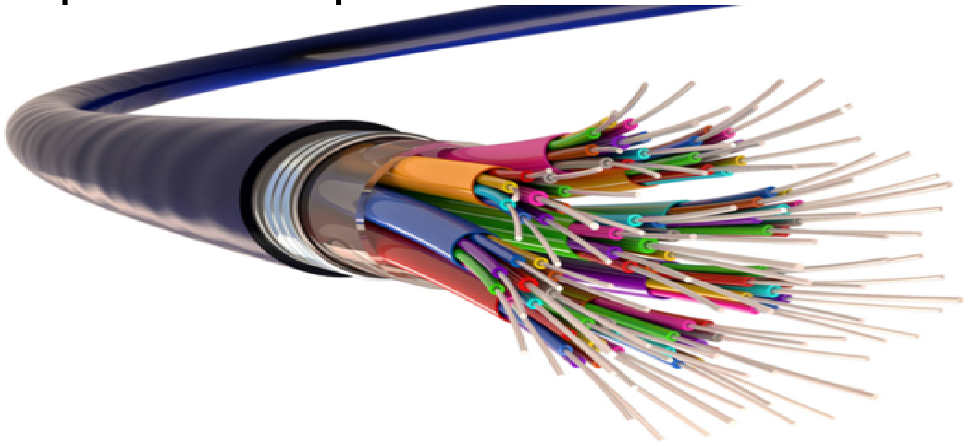
# Coaxial Cable

- Consists of two concentric copper conductors.
- Bidirectional
- Widely used in cable television systems.
- Can serve as a guided shared medium
- Multiple end systems connected directly to the cable can receive signals from each other.



# Fiber Optics

- Glass fiber carrying light pulses, each pulse represents a bit.
- Thin and flexible medium.
- High speed operation
- Supports high bit rates
- Immune to electromagnetic interfer
- Preferred for long-distance guided transmission



# Satellite Radio Channels

- Links ground stations using microwave transmitters/receivers
- Two types: Geostationary
  Low-Earth Orbiting (LEO)
- Geostationary Satellites:
- Permanently positioned above a fixed spot on Earth.
- Ideal for areas without DSL or cable-based Internet access.
- Low-Earth Orbiting (LEO)
- Closer to Earth, no fixed position

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# Layered Architecture

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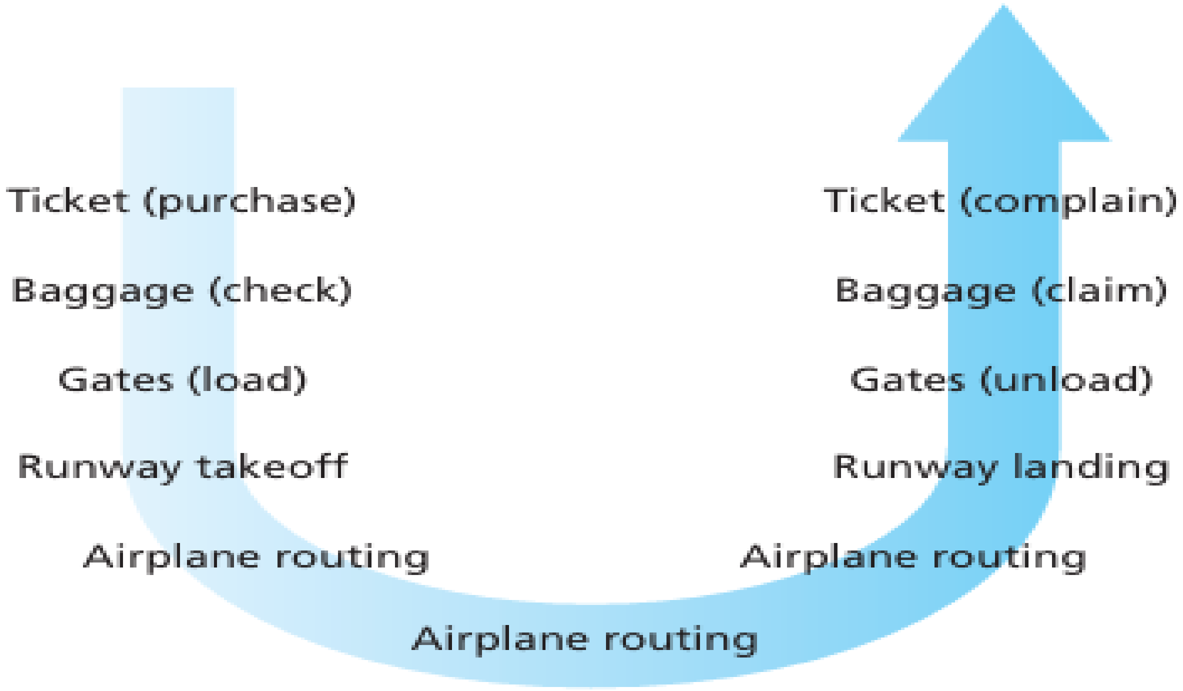
- Protocol Layering
- Internet Protocol Stack
- OSI Reference Model

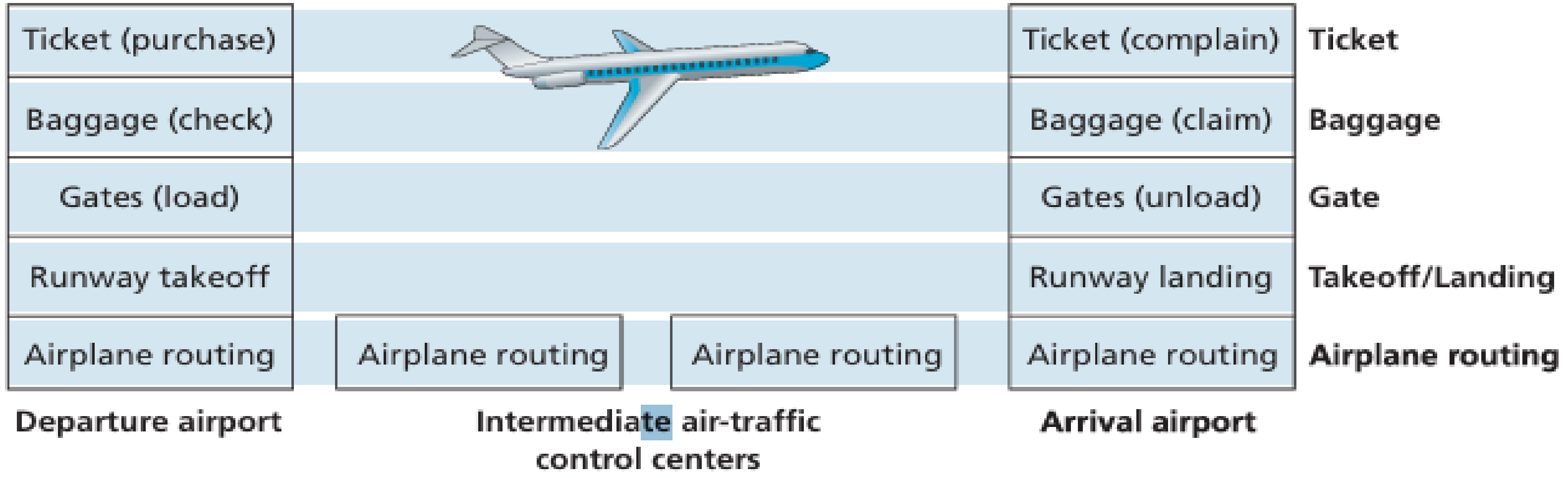
# Organization of Air Travel

- A series of steps



# Layering of Airline Functionality

- The functionality of the airline system is Horizontally layered.
- Each layer performs specific functions or services.
- The functionality is distributed across layers in a modular fashion



- Layers utilize services from the layer directly below.
- Enables a clear and modular organization of system functionalities.
- Changes in one layer do not disrupt the entire system.

# Protocol Layering

- Network protocols are structured in layers for organized design.
- Each layer has specific functions and interacts with adjacent layers.
- Protocols can be implemented in software, hardware, or a combination
- Layering offers conceptual clarity and structural organization.
- Drawbacks: Duplication of functionality between layers.
  Challenges in information sharing between layers.

# Internet Protocol Stack

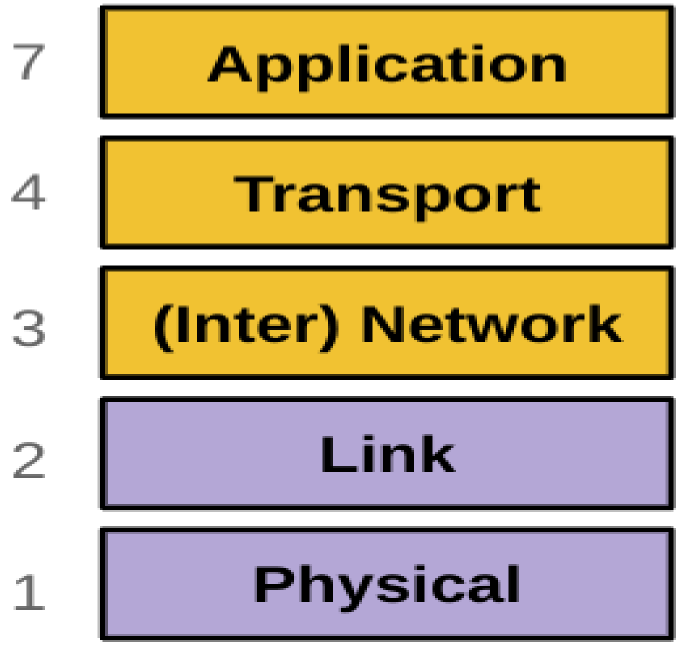
- Protocols from various layers form the Protocol stack.
- Internet protocol stack comprises physical, link, network, transport, and application layers.

Application layer :

- Hosts network applications and their proto
- Eg: HTTP, SMTP, FTP and DNS

Transport Layer

- Transports application-layer messages
- Protocols: TCP, UDP



- TCP provides guaranteed delivery, flow control, and congestion control.
- Transport-layer packets referred to as segments.

## Network Layer

- Moves datagrams between hosts.
- Receives transport-layer segment and destination address
- Often referred to as the IP layer.
- Eg:IP Protocol and various routing protocols

## Link Layer

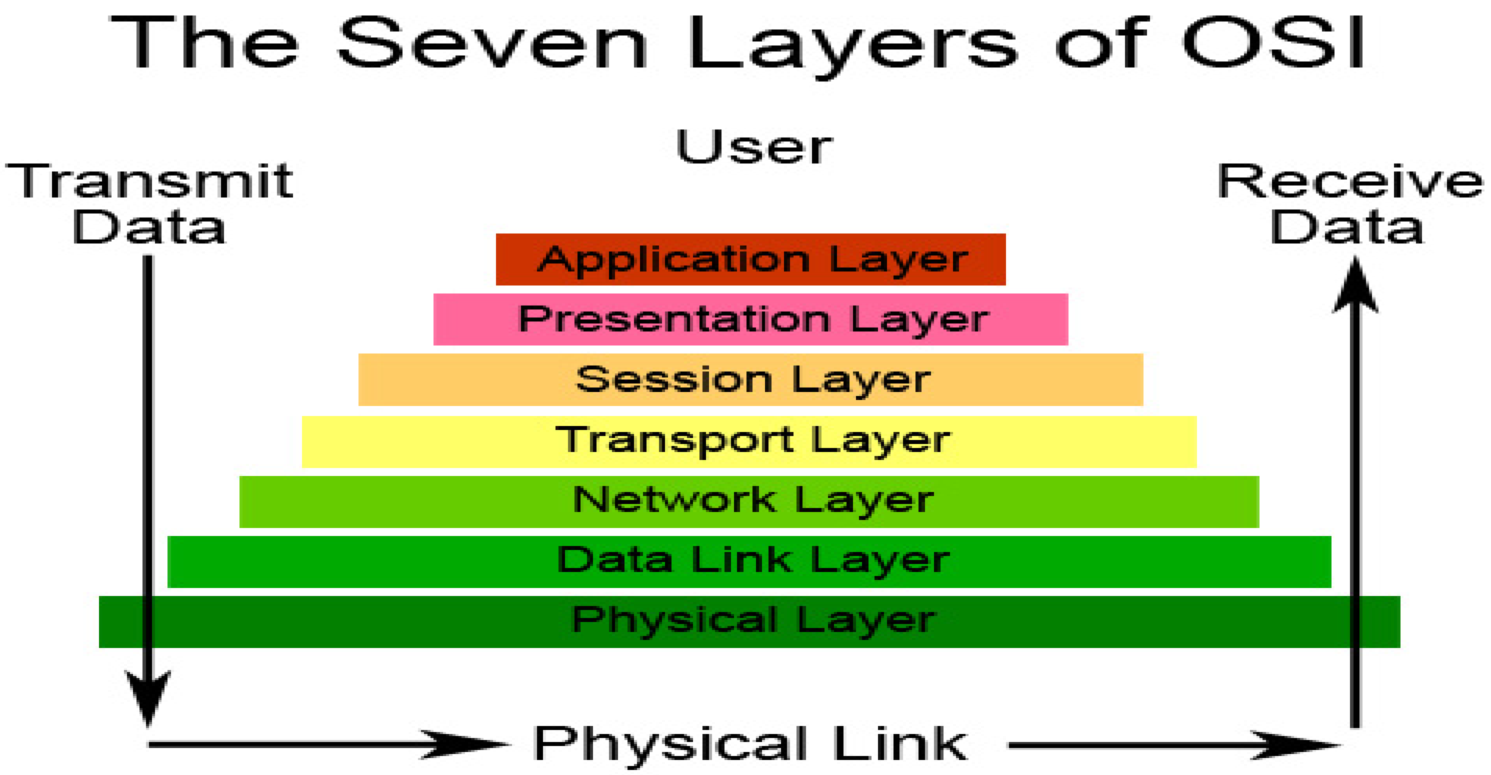
- Routes datagrams through routers via the link layer.
- Link layer services utilized at each node along the route.
- Different link-layer protocols can handle a datagram at various links.

Eg: Ethernet, 802.111 (WiFi), PPP

## Physical Layers

- Moves individual bits within frames from one node to another
- Link-dependent protocols, varying with the transmission medium

# OSI Reference Model



# ISO/OSI Reference Model

- OSI model was proposed by the International Organization for Standardization (ISO) .
- Consists of seven layers, serves as a conceptual framework for designing computer networks.

Seven Layers:

- Application Layer
- Presentation Layer
- Session Layer
- Transport Layer
- Network Layer
- Data Link Layer
- Physical Layer

- IP stack consists of five layers, omits the Presentation and Session layers

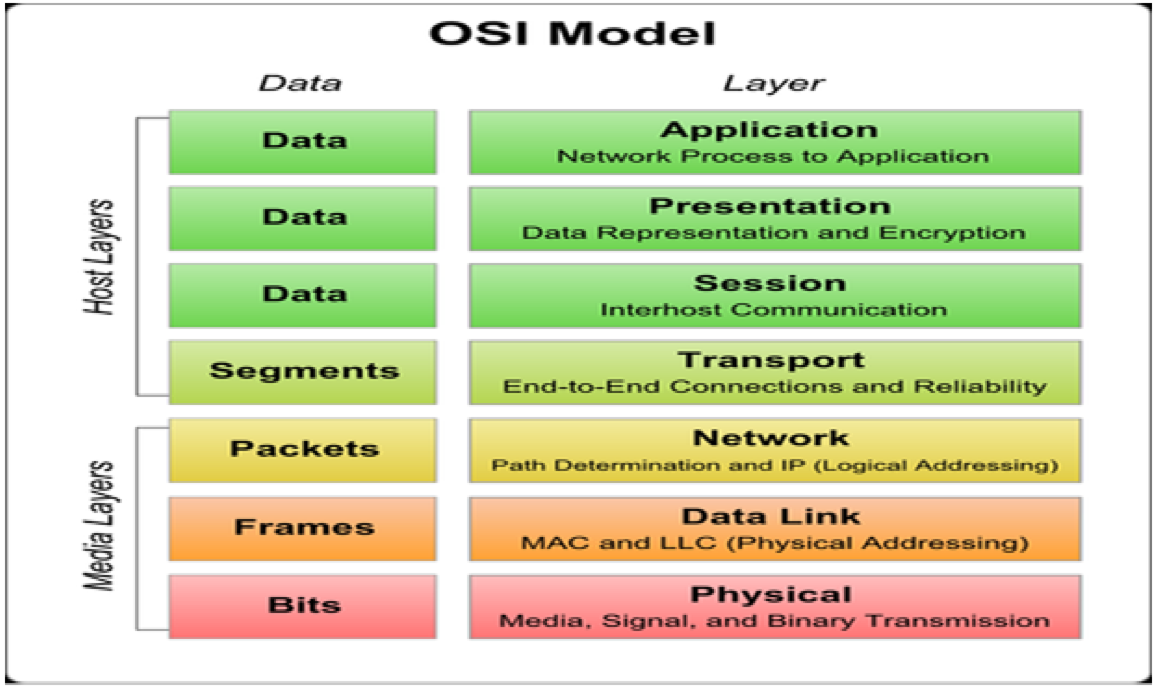
## Presentation Layer:

- Provides services for interpreting the meaning of exchanged data.
- Services include data compression, data encryption, and data description.

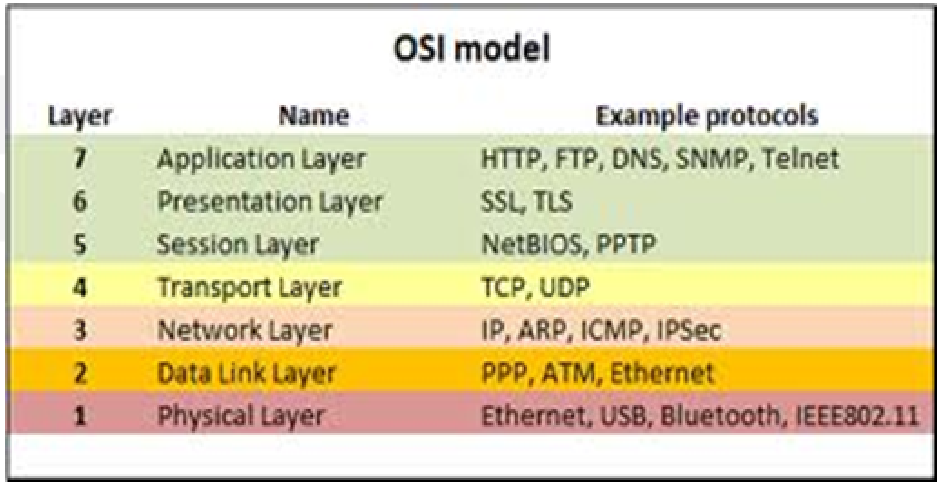
## Session Layer:

- Manages delimiting and synchronization of data exchange.
- Includes support for building checkpointing and recovery schemes.

- Facilitates reliable and organized communication between applications.





OSI model

| Layer | Name | Example protocols |
|---|---|---|
| 7 | Application Layer | HTTP, FTP, DNS, SNMP, Telnet |
| 6 | Presentation Layer | SSL, TLS |
| 5 | Session Layer | NetBIOS, PPTP |
| 4 | Transport Layer | TCP, UDP |
| 3 | Network Layer | IP, ARP, ICMP, IPSec |
| 2 | Data Link Layer | PPP, ATM, Ethernet |
| 1 | Physical Layer | Ethernet, USB, Bluetooth, IEEE802.11 |

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# Data Transfer Methods

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- Data Transfer Methods :
- Packet Switching
- Circuit-Switching
- Overview of Delay
- Loss and Throughput

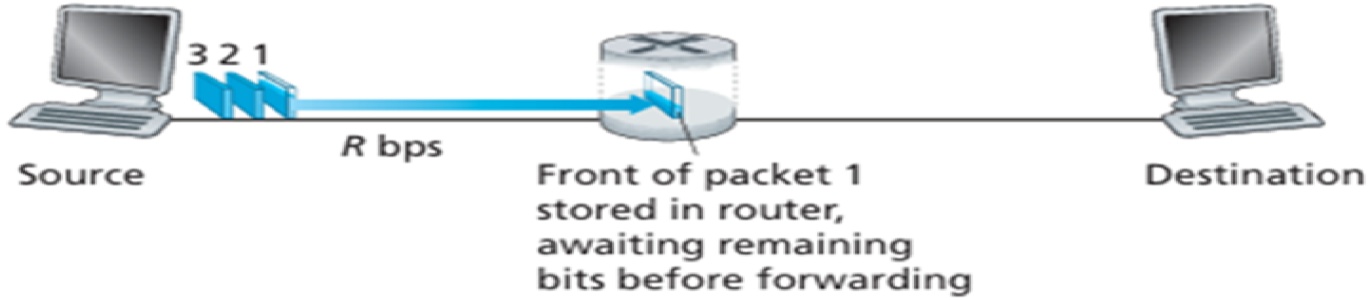
# Packet Switching

- A Fundamental method for data transfer in networks.
- End Systems exchange messages in a Network application.
- Long messages are broken into smaller units called Packets.
- Packets travel through communication links and packet switches.
- If a packet of L bits is sent over a link with transmission rate R bits/sec:

  Time to transmit the packet = L/R seconds

# Store-and-Forward Transmission

- Most packet switches employ Store-and-forward Transmission.
- Packet switch must receive the complete packet before transmission.

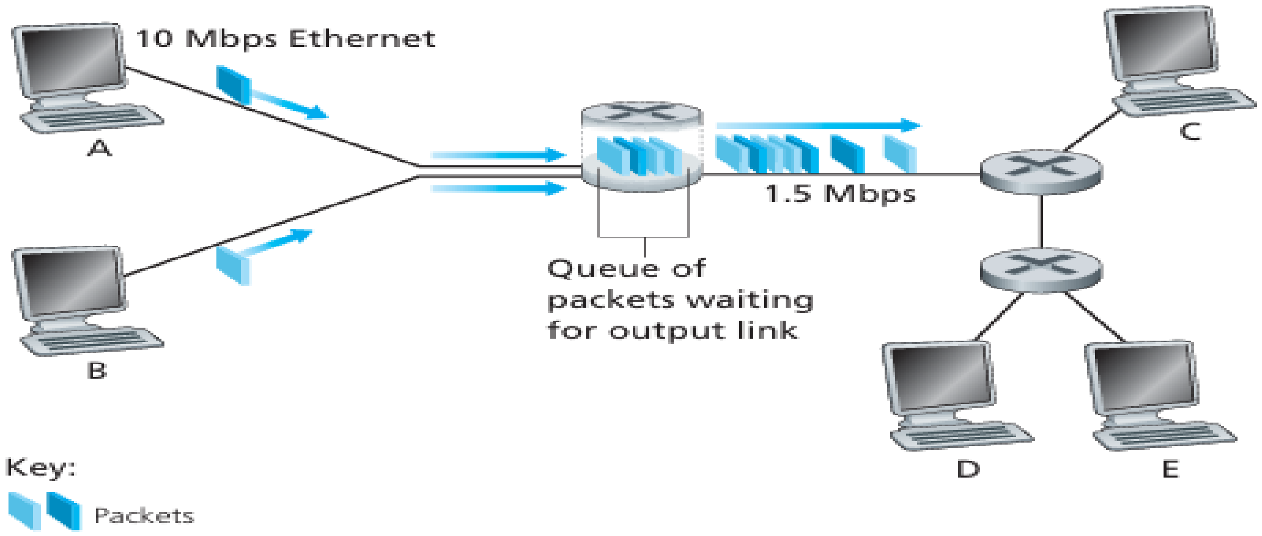



- End-to-End delay for sending one packet over N links.

End-to-end delay = N * (L/R)

# Queuing Delays and Packet Loss

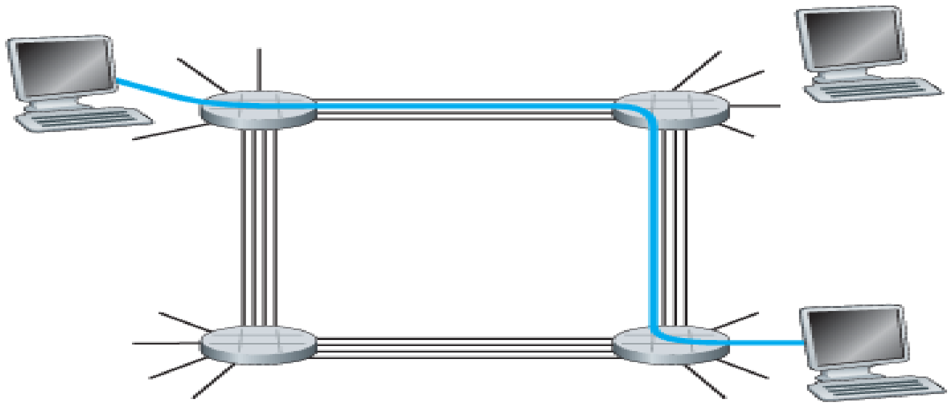
- Each packet switch has output buffers for attached links.
- Output buffers store packets waiting for transmission.
- Handle situations where a link is busy transmitting another packet.



- If arrival rate exceeds link’s transmission rate, congestion occurs.
- Packets queue in the link's output buffer, waiting for their turn to be transmitted.
- Risk of losing packets if the buffer memory becomes full.

# Circuit-Switching

- Circuit-switched networks reserve resources for the entire communication session.
- Resources are pre-allocated, ensuring dedicated communication.
- Commonly used in traditional telephone network.



- Hosts connect directly to switches within the network.
- Enable communication by establishing dedicated end-to-end connections through circuit reservation.
- Offers guaranteed transmission.

# Packet Switching Versus Circuit Switching

| | Packet Switching | Circuit Switching |
|---|---|---|
| **Resource Allocation** | Dynamically allocated based on demand | Pre-allocated resources for the entire session |
| **Transmission Capacity** | Shared dynamically among users | Dedicated link capacity per user for the entire session. |
| **Efficiency** | Efficient link capacity utilization | Less efficient |
| **Delay** | Variable end-to-end delays due to queuing | Predictable and constant end-to-end delays |
| **Application** | Suited for non-real-time services | Suited for real-time services |

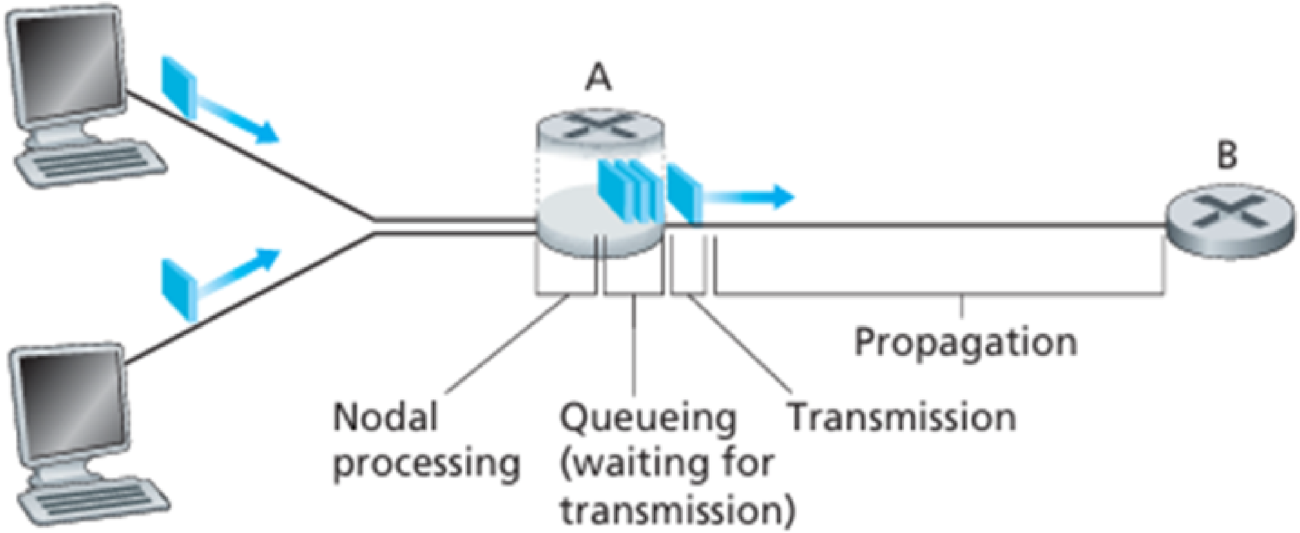
# Delay

- Packet Starts in host, passes through routers, ends in another host.
- Delays occur at each node during the packet's journey.

Types of Delay:

- Nodal processing delay
- Queuing delay
- Transmission delay
- Propagation delay

# Types of Delay




End-to-End Delay, dend-end = N (dproc + dtrans + dprop)

# Sources of Packet Delay

## Processing Delay

- Time to examine packet's header, determine outbound link, and check for bit-level errors.
- Typically microseconds or less in high-speed routers

## Queuing Delay

- Packet waits in queue, duration depends on queued packets.
- Range from microseconds to milliseconds.

## Transmission Delay

- Time to transmit packet's bits into the link.
- Typically in microseconds to milliseconds.

- Packet length (L bits) and transmission rate (R bits/sec)

Transmission delay = L/R

## Propagation Delay

- Time for a bit to propagate from link beginning to next node
- Depends on physical medium, typically milliseconds.
- Distance between routers (d) and propagation speed (s)

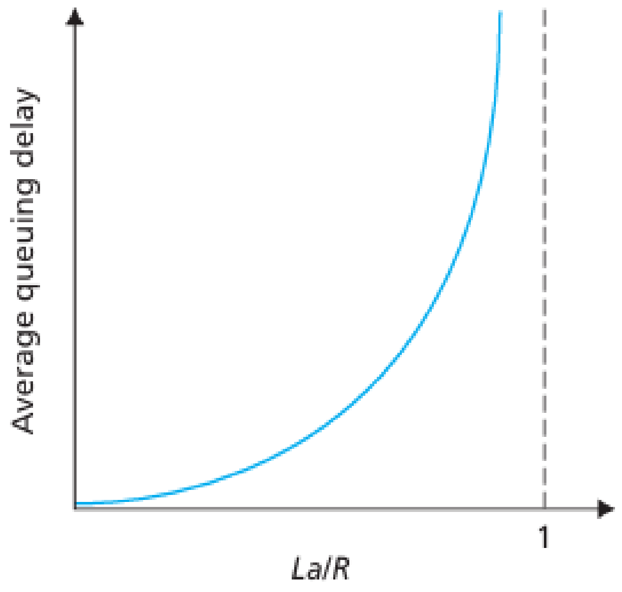
Propagation delay = d/s

# Queuing delay and Packet Loss

- Queuing delay varies from packet to packet.
- *R*: link bandwidth (bps), *L*: packet length (bits) and a: average packet arrival rate

Traffic intensity= *La/R*

- La/R ~ 0: avg. queueing delay small
- La/R -> 1: avg. queueing delay large
- La/R > 1:queuing delay approach infinity



# Packet Loss

- Queues have finite capacities.
- Finite queue capacity prevents packet delays from approaching infinity
- When a packet arrives at a full queue, the router drops the packet, resulting in packet loss.
- Lost packet may be retransmitted by previous node by source end system.

# Throughput

- Rate (bits/time) at which bits transferred between sender/receiver.
- Instantaneous throughput : Rate at given point in time
- Average throughput : Rate over longer period of time

Average throughput = F/T bits/sec

- Different applications have specific throughput requirements.

# File Transfer Throughput

- Two-Link Network: Throughput determined by the minimum rate between server-router (Rs) and router-client (Rc)
- Throughput = min{Rs, Rc}
- File transfer time: F / min{Rs, Rc}
- Network with N Links:min{R1, R2, ..., RN}

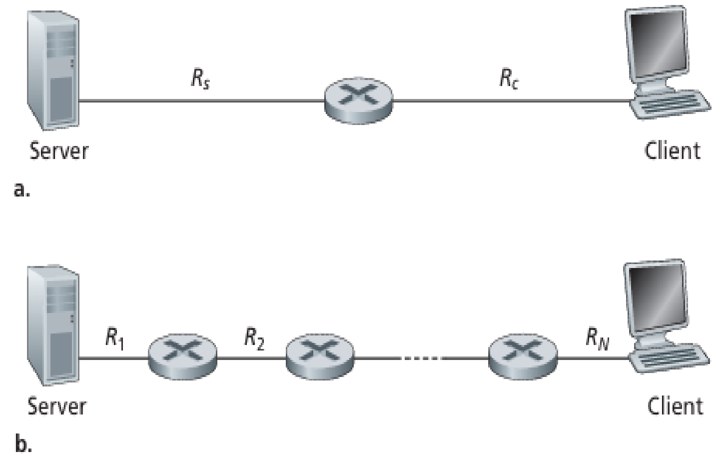



Fig : Throughput for a file transfer from server to client